Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/20/2026
PI Name:	Crowe, Kelly Ph.D.		
Project Title:	Assessment of Sialylation in Skeletal Muscle Atrophy Due to Simulated Microgravity		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition (2) **Muscle**:Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	45207-1035	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2023 HERO NNJ23ZSA001N-OMNIBUS : NASA Human Research Program Omnibus Opportunity
Start Date:	01/01/2024	**End Date:**	12/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per NSSC information (Ed., 11/27/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mortreux, Marie Ph.D. (University of Rhode Island)		
Grant/Contract No.:	80NSSC24K0434		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Long-term skeletal muscle unloading due to microgravity exposure in spaceflight conditions has a striking effect on the health of skeletal muscle, which may impact performance of flight mission tasks. While resistance exercise has shown efficacy as a countermeasure, capacity constraints may preclude effective implementation of these exercise protocols on the lunar surface. As such, it is crucial to identify novel targets to ameliorate skeletal muscle atrophy. One such target is sialic acid (SA), a terminal glycan on extracellular glycan chains, as changes in SA are associated with skeletal muscle weakness and wasting due to their role in development, regeneration, and contractility. In the rare muscle disease GNE myopathy, a reduction in SA levels leads to progressive muscle weakness and wasting. Furthermore, SA alterations have been seen in atrophy; for example, expression of sialidases, which remove SA from glycoproteins, have shown alterations in a model of atrophy. While global perturbations in SA levels certainly impair skeletal muscle function, the linkage of SA appears to mediate these effects; in skeletal muscle of patients with GNE myopathy, hyposialylation of O-linked, but not N-linked glycans, was correlated to pathology. We hypothesize that simulated microgravity will lead to linkage-specific alterations in skeletal muscle sialylation mediated by changes in sialidases and/or sialyltransferases. To address this, we will use immunofluorescence with a panel of SA-detecting lectins, glycanbinding proteins, as well as lectin microarrays, to probe alterations in linkage-specific SA abundance in skeletal muscle tissue from rats with or without a hindlimb-unloading protocol. Next, we will probe expression of the sialyltransferases that form SA linkages and the sialidases that enzymatically remove them, via qRT-PCR and Western blotting to assess the underlying mechanism of these SA alterations and identify novel pharmacological targets. This will be the first study to address alterations in skeletal muscle sialylation in the context of spaceflight-induced muscle atrophy. This would inform future studies to assess rescue of sialylation in skeletal muscle as a treatment for atrophy, as skeletal muscle sialylation can be altered via oral supplementation and gene therapy; in fact, sialylation-altering nutraceutical therapies for skeletal muscle disease have recently progressed through clinical trials for the rare disease GNE myopathy. This work aims to develop novel targets to address skeletal muscle atrophy via nutraceutical and pharmacological approaches, which would help to ensure effective performance of flight mission tasks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This will be the first study to address alterations in skeletal muscle sialylation in the context of spaceflight-induced muscle atrophy. This would inform future studies to assess rescue of sialylation in skeletal muscle as a treatment for atrophy, as skeletal muscle sialylation can be altered via oral supplementation and gene therapy; in fact, several glycan-altering therapies for skeletal muscle disease have recently progressed through clinical trials. This work represents a novel target to address spaceflight-relevant skeletal muscle atrophy, which could potentially help to inform treatment for other types of skeletal muscle atrophy such as sarcopenia and cachexia.
Task Progress:	Specific Aim 1: Determine linkage-specific skeletal muscle sialylation in a rodent hindlimb-suspension model to simulate microgravity exposure. The goal of this aim was to use both lectin staining and lectin microarray to look at changes in sialic acid abundance in both male and female rats with and without a hindlimb-unloading protocol for 14 days to induce unloading atrophy. Both gastrocnemius (a predominantly fast-twitch muscle) and soleus (a predominantly slow-twitch muscle) were used for further analysis. Lectin Staining: We stained the gastrocnemius and soleus for all hindlimb-unloading and control rats with several lectins. We used sialidase, an enzyme that removes sialic acid, as a control for sialylation-detecting lectins; only those lectins in which we were able to distinguish between sialidase treated and untreated samples were used for further analysis. Based on this criterion, we collected images using time-matched fluorescent microscopy for four lectins: Wheat Germ Agglutinin (WGA, recognizing any bound SA), Sambucus nigra agglutinin (SNA, recognizing Siaα2-6Gal/GalNAc), and Maackia amurensis agglutinin (MAA, recognizing Siaα2-3Gal). This data is a qualitative measure of lectin binding to validate lectin microarray data. Lectin Microarray: Lectin microarray was performed in gastrocnemius and soleus of hindlimb-unloaded and control rats. In male rats' gastrocnemius muscle, hindlimb unloading was associated with decreased core O-glycan and 2,3-sialic acid binding, compared against normal loading. Similarly, in male rats' soleus muscle, hindlimb unloading was associated with the same decreases in binding, though the difference in core O-glycan binding was non-statistical. In female rats' gastrocnemius muscle, hindlimb unloading was associated with decreased polyLacNAc binding compared against normal loading. In female rats' soleus muscle, hindlimb unloading was associated with increased core O-glycan and sialic acid binding, compared against normal loading. These differences were similar to those seen in male rats relative to female under normal loading, implying the females' glycans post-unloading were meeting the males' levels. Specific Aim 2: Assess expression of sialyltransferases and sialidases in skeletal muscle of a rodent hindlimb-suspension model. The goal of this aim was to probe changes in sialidases and sialyltransferases, enzymes that could be responsible for altering sialic acid abundance in both male and female rats with and without a hindlimb-unloading protocol for 14 days to induce unloading atrophy. While we had initially planned to use a singleplex approach, in an effort to preserve sample, we have decided to instead employ a duplex method. In this method, we use a Taq DNA polymerase-based assay with VIC and FAM fluorescent reporter dyes for the endogenous control and gene of interest respectively. This allows for simultaneous quantification of our gene target and housekeeping gene in a single well, which halves the amount of sample needed to quantify all of our targets. This is particularly important for the soleus muscle, which was quite limited. After trialing both GAPDH and 18s ribosomal RNA as endogenous control genes, we selected 18s. We validated our duplex approach for all targets in rat skeletal muscle samples by demonstrating that the amplification of both the 18s endogenous control and our genes of interest was acceptably similar when run in singleplex and duplex. We collected data on expression of several key genes that encode enzymes that add or remove sialic acid to glycan chains for hindlimb-unloaded and control rats. There are several mammalian STs, each of which catalyzes different attachment of SA to the underlying glycans, with α2,3, α2,6, and α2,8 attachments catalyzed by ST3, ST6, and ST8, respectively (each with multiple isoforms with varying tissue expression). Similarly, SAs are enzymatically removed in a

	linkage-specific manner by four specific mammalian sialidases, designated NEU1, NEU2, NEU3, and NEU4. Of these, a subset was chosen for analysis based on reported tissue expression in mouse skeletal muscle: NEU1, ST3GAL1, ST3GAL2, ST3GAL5, ST8SIA2, and ST8SIA4. Gene expressions (2-ΔΔCT) were compared between hindlimb suspension and control animals. Of these, ST3GAL2 showed significantly higher expression in the NL group ($p = 0.03$). How the results have been disseminated: As it was near the beginning of the granting period, preliminary data and grant plans were presented at the 2024 NASA Human Research Program (HRP) Investigators' Workshop (IWS). Initial data collected during the granting period was presented at the American Society for Gravitational and Space Research (ASGSR) 2024 meeting and the 2025 HRP IWS, and final data will be submitted to the 2026 HRP IWS. Results from this work are in preparation as a manuscript to be submitted in a peer-reviewed journal. Presented and/or abstract submitted: Mackey, B., Mazza, N., Cullen, R., Mortreux, M., & Crowe, K.E. Skeletal Muscle Sialylation in Simulated Microgravity. Poster presentation delivered at the Human Research Program Investigators' Workshop (HRP IWS), February 2024. Green, T., Greulach, E., Hornberger, C., Leontescu, N., Mathis, J. Parker, O., Simone, E., Vonderhaar, R.,Woods, J., Mortreux, M., Wetzel, H., Crowe, K. E. Characterization of Skeletal Muscle Sialylation in a GroundBased Rat Model of Microgravity. Poster presentation delivered at the American Society for Gravitational and Space Research's (ASGSR) Annual Meeting, December 2024. Greulach, E., Hornberger, C., Leontescu, N., Mathis, J. Parker, O., Simone, E., Vonderhaar, R., Woods, J., Mortreux, M., Wetzel, H., Crowe, K. E. Assessment of sialylation in skeletal muscle of a ground-based rat model of microgravity. Poster presentation delivered at the Human Research Program Investigators' Workshop (HRP IWS), January 2025. Borhorst, J., Cagle, S., Linnemann, E., Thomas, T., Mortreux, M., Wetzel, H., Crowe, K. E. Effects of skeletal muscle unloading on glycan dynamics. Abstract in preparation for the Human Research Program Investigators' Workshop (HRP IWS), February 2026. [Ed. Note: See Bibliography for additional information.]
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Mackey B, Mazza N, Cullen R, Mortreux M, Crowe KE. "Skeletal muscle sialylation in simulated microgravity. " 2024 Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Poster. Abstracts. 2024 Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Green T, Greulach E, Hornberger C, Leontescu N, Mathis J, Parker O, Simone E, Vonderhaar R, Woods J, Mortreux M, Wetzel H, Crowe KE. "Characterization of skeletal muscle sialylation in a ground-based rat model of microgravity. " 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Poster. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Greulach E, Hornberger C, Leontescu N, Mathis J, Parker O, Simone E, Vonderhaar R, Woods J, Mortreux M, Wetzel H, Crowe KE. "Assessment of sialylation in skeletal muscle of a ground-based rat model of microgravity." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Poster. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025